

CONSERVATION OF MOMENTUM

In any collision or interaction between two or more objects in an isolated system, the total momentum of the system will remain constant.

i.e. The total initial momentum = the total final momentum.

This can be expressed as follows.

$$\Sigma p_i = \Sigma p_f$$

where Σp_i = total initial momentum of all parts of the system
 Σp_f = total final momentum of all parts of the system

NOTE

An isolated system is one on which no external unbalanced force is acting. This is very difficult to achieve on Earth because of gravity, friction and air resistance.

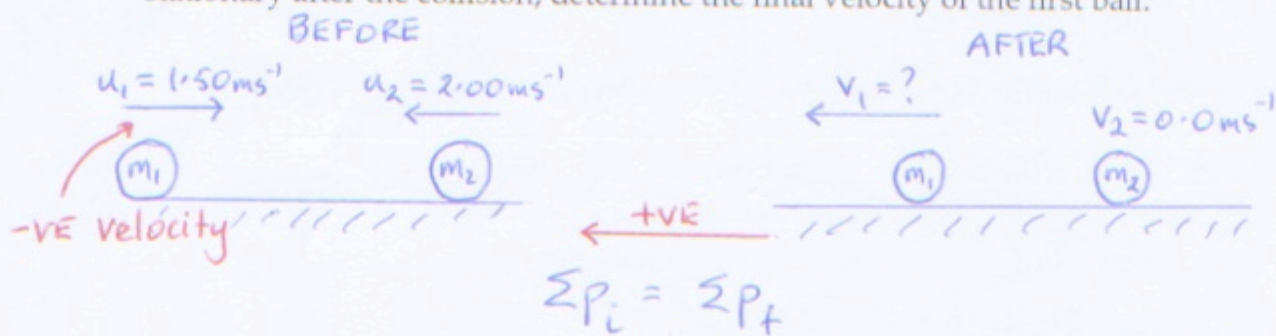
For the purposes of calculations, **it will be assumed that all systems are isolated** and that the only forces acting are the action/reaction forces during collision.

Also for the ease of calculation, **only problems involving one dimension will be considered**, though it is relatively easy to consider vector additions in two dimensions.

EXAMPLES

1. COLLISIONS

A billiard ball of mass 0.200 kg moving at 1.50 ms^{-1} collides head-on with a similar ball moving at 2.00 ms^{-1} in the opposite direction. If the second ball becomes stationary after the collision, determine the final velocity of the first ball.



$$\Rightarrow m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

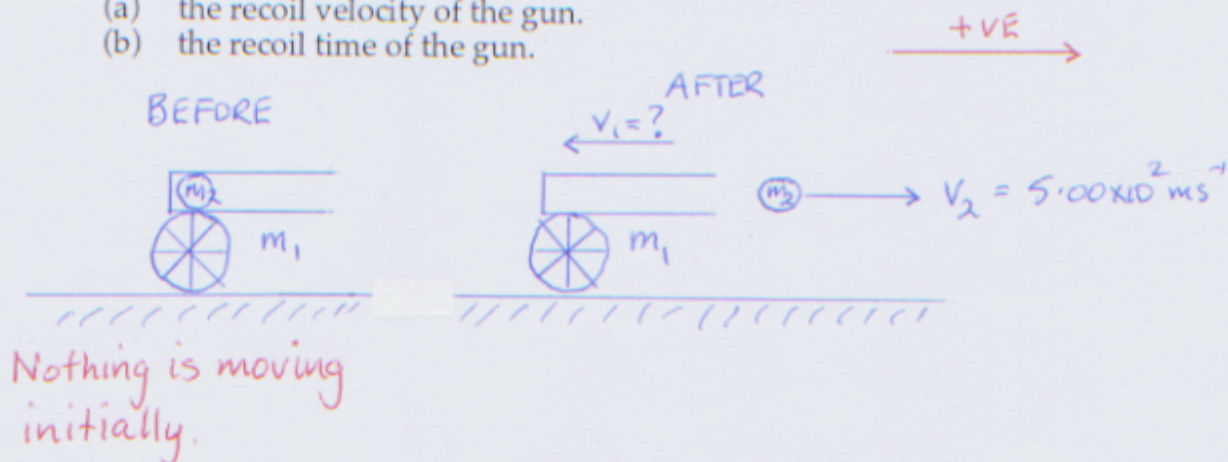
$$\Rightarrow (0.200)(-1.50) + (0.200)(2.00) = (0.200)v_1 + 0$$

$$\Rightarrow \underline{v_1 = 0.500 \text{ ms}^{-1} \text{ backwards.}}$$

2. EXPLOSIONS

A 9.50×10^2 kg gun fires a 2.00 kg shell with a muzzle velocity of 5.00×10^2 ms⁻¹. The gun recoils against a resisting force of 5.00×10^3 N. Calculate:

- (a) the recoil velocity of the gun.
 (b) the recoil time of the gun.



(a) $\Sigma p_i = \Sigma p_f$

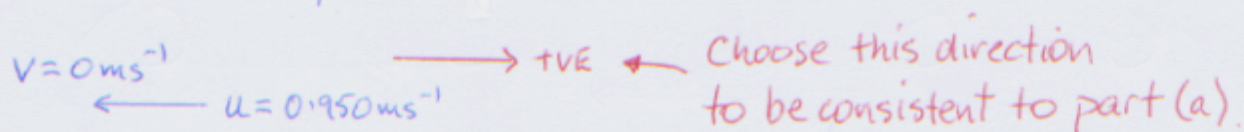
$$\Rightarrow m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$\Rightarrow 0 + 0 = (9.50 \times 10^2) v_1 + (2.00)(5.00 \times 10^2)$$

$$\Rightarrow v_1 = -0.950 \text{ ms}^{-1}$$

$\therefore$ Recoil velocity = 0.950 ms^{-1} backwards.

- (b). The stopping of the gun is considered as a separate situation.



Must calculate Δv first.

$$\Delta v = v - u$$

$$= 0 - (-0.950)$$

$$= 0.950 \text{ ms}^{-1} \text{ in the direction of the shell,}$$

$$I = Ft = m\Delta v = \Delta p$$

$$\Rightarrow t = \frac{m\Delta v}{F} = \frac{(9.50 \times 10^2)(0.950)}{5.00 \times 10^3}$$

$$= \underline{0.180 \text{ s}}$$